



ADAM

AI-Driven Air Monitoring



Move. Measure. Change.

Cities in motion. Air under control.

We turned the city into a sensor.

The Problem

You can't fix what you can't measure

Air pollution is the top environmental health risk in Europe, yet most cities lack the infrastructure to monitor it properly. Existing networks of stationary stations cover only 5–10% of urban areas, leaving critical data gaps that prevent effective action.

Why This Matters Now

- 330,000+** annual premature deaths in Europe are linked directly to air pollution (EEA, 2023)
- 95%** of urban Europeans breathe air that exceeds WHO pollution guidelines
- 50,000** companies are now legally required under the EU CSRD directive (from 2025) to report verified environmental data including air quality — or face fines
- 2024** brings stricter EU air quality limits, pushing many cities out of compliance
- €50,000–€150,000** is the cost of just one stationary monitoring station, making dense networks financially impossible



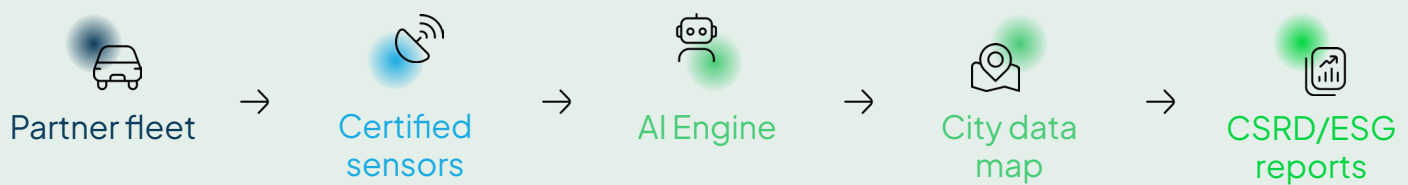
The Solution

What is ADAM?

ADAM (AI-Driven Air Monitoring) is designed to install compact sensor modules on the rooftops of partner vehicles — taxis, delivery and logistics fleets. As these vehicles move through the city, they continuously collect real-time air quality data at street level, building a dynamic pollution map that stationary stations simply cannot provide.



How It Works



AI Engine

Over 100,000 measurements per vehicle per year — processed through algorithmic validation combined with AI-assisted analysis and forecasting in real time:

01 — Data Validation

Filters dirty readings from tunnels, parking and heavy traffic. Cross-checks with nearby certified stations. Labels data by validity level.

02 — City Analysis

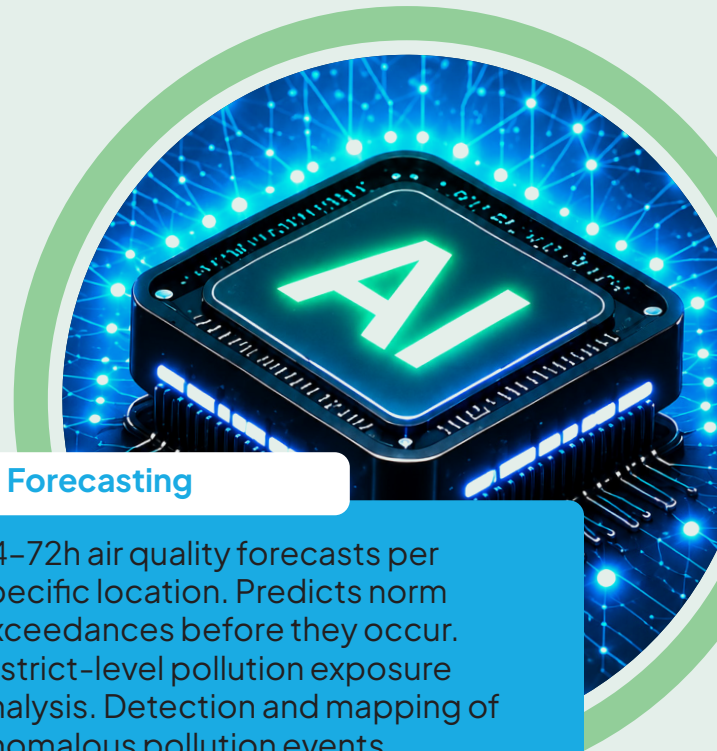
Identifies pollution hotspots and recurring patterns. Correlates data with meteorological parameters (wind, humidity, temperature). Detects morning peaks, seasonality, day/night cycles. Heatmap updated every 5–15 minutes. Street-level simulation is a long-term development goal following first large-scale deployment.

03 — Forecasting

24–72h air quality forecasts per specific location. Predicts norm exceedances before they occur. District-level pollution exposure analysis. Detection and mapping of anomalous pollution events.

04 — Recommendations & Optimization

Data for optimal placement of stationary sensors. Traffic rerouting suggestions to reduce pollution. Urban planning inputs: schools, parks, offices. Auto-generated CSRD/ESRS compliance reports.

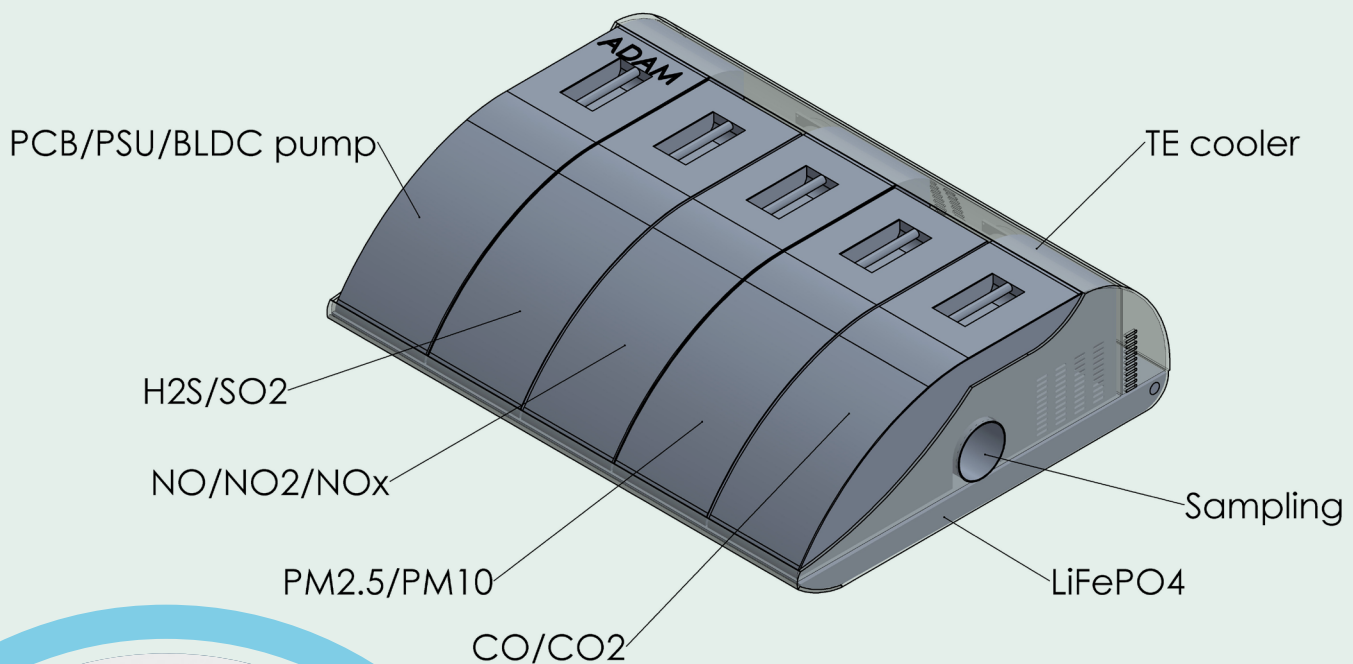


Our Product

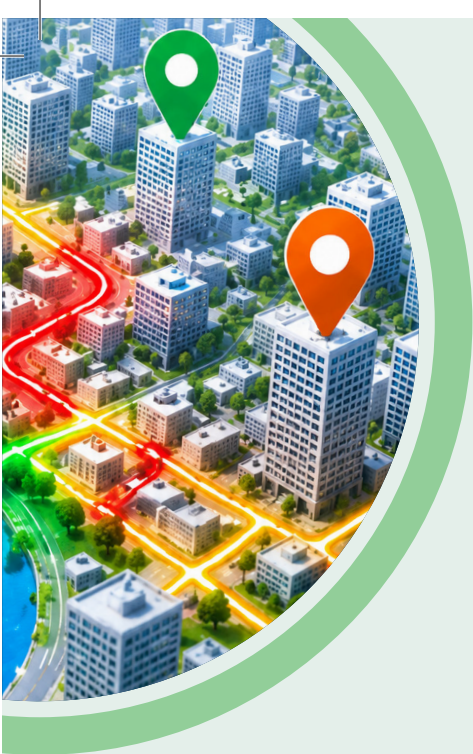
Hardware · Software · Platform

Hardware — Mobile Sensor Modules

- **Modular plug-and-play design** — compact, lightweight, easy rooftop installation on partner vehicles
- **NDIR method** — CO/CO₂ concentration measurements
- **Chemiluminescence** — NO/NO₂/NO_x/NH₃/O₃ measurements
- **UV-fluorescence** — SO₂/H₂S/TRS measurements
- **Optical counter** — PM1/PM2.5/PM5/PM10 and TSP dust monitoring
- **Electrochemical sensors** — additional pollutant measurement
- **Li-Ion accumulator** — 24/7 uninterrupted operation when vehicle power is unavailable
- **Thermoelectric coolers** — high-precision temperature stabilisation
- **EHD-based air amplifier** — continuous high-volume air sampling



During operation, air is continuously sampled from outside, heated to evaporate moisture and stabilise temperature, then measured simultaneously across all modules. Targeting EN 15267-4 certification — data legally recognised for EU regulatory reporting and ESG audits.



Software — Platform

- **Real-time city air quality dashboard** — street-level heatmap updated every 5–15 minutes
- **Fleet tracking** — live positions of all monitoring vehicles
- **Automated alerts** — notifications when pollutant norms are exceeded
- **Automated CSRD/ESRS E1/E2 compliance report generation** — PDF and Excel export
- **ESG building passport** — for BREEAM, LEED, EU Taxonomy and green financing
- **API access** — integration into client systems, navigation platforms and insurance tools
- **Mobile app (B2C)** — personal air quality alerts, clean air route planning, pollution exposure insights



Every Sector. One Solution.

ADAM addresses real challenges across every sector that depends on verified air quality data

Municipalities & City Administrations

B2G

CHALLENGE

Cities must monitor air quality across their entire territory — but stationary stations cover only 5–10% of urban areas and cost €50,000–€150,000 each.

SOLUTION

Real-time air quality map of every street via partner fleets. Automated EU Air Quality Directive compliance reporting. Data for traffic management and urban planning.

Industrial Enterprises & Logistics

B2B

CHALLENGE

Under EU CSRD (mandatory from 2025), 50,000 companies must submit verified air quality data — most have no way to collect it.

SOLUTION

Continuous monitoring around facilities and along fleet routes. Auto-generated CSRD/ESRS E1/E2 compliant reports ready for regulators and auditors.

Real Estate Developers

B2B

CHALLENGE

ESG certification and green financing require verified environmental data for specific locations — often unavailable or expensive to obtain.

SOLUTION

ESG building passport — a verified air quality profile for any location, supporting BREEAM, LEED and EU Taxonomy certification.

Citizens

B2C

CHALLENGE

No reliable way to know real air quality on your street, running route or near your child's school.

SOLUTION

Mobile app with real-time alerts, clean air routes and personal pollution exposure insights.

Why ADAM

*The only mobile air monitoring system
targeting EN 15267-4 certification*

Competitive Advantages

Full city
coverage

Vehicles cover 100% of city streets — not just 10 fixed points. Every street, every neighbourhood, every hour.

AI-validated
data

Up to 96% accuracy on validated direct measurements. Over 100,000 measurements per vehicle per year continuously improve our models.

CSRD-ready
reports

Auto-generated ESRS E1/E2 compliant reports accepted by EU regulators, auditors and banks. One click — the report is ready.

10× more
affordable

One stationary station costs €50,000–€150,000. ADAM covers an entire city via partner fleets at a fraction of that cost.

EN 15267-4
certification

Targeting the only standard that makes mobile air quality data legally recognised for regulatory reporting and ESG audits.

Asset-light
model

No proprietary fleet required. We partner with existing taxi and logistics operators — rapid deployment, low capital cost.



About Us

SINTROL LLC is an advanced technology company based in Kyiv, Ukraine, specialising in the development of monitoring systems for gases, dust and air quality. ADAM is built on SINTROL's engineering expertise in industrial environmental monitoring.

We offer a wide range of services including development of systems according to customer technical requirements, documentation preparation, development of specialised software for system control, supply, installation, warranty and post-warranty support.



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